



BRAC UNIVERSITY
Department of Computer Science & Engineering
FINAL EXAMINATION, Spring 2025
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 60 (20×3)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.

Name:	ID:	Section:
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Question 1

[20]

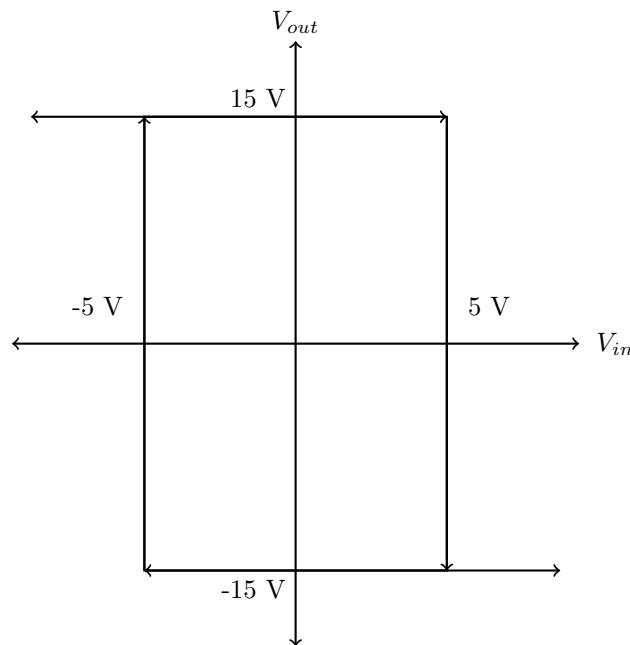


Figure 1: VTC of a Schmitt Trigger

CO1	(a)	What <i>type</i> of Schmitt Trigger is shown in Figure 1? Find the hysteresis width, shift voltage (V_s), bias voltages (V_H , V_L), threshold voltages (V_{TH} , V_{TL}) for the given Schmitt Trigger.	[1+2]
CO1	(b)	Design a Schmitt Trigger Circuit that will produce the same VTC.	[5]
CO1	(c)	Modify your designed Schmitt Trigger in (b), so that the <i>shift voltage</i> becomes +10 V. Draw the VTC and the modified circuit for this case.	[6]
CO1	(d)	Apply an input signal $V_{in}(t) = -12\sin(2\pi ft)$ to the Schmitt Trigger & VTC derived in (c) and draw the output V_{out} . [Show the output for at least one period of the input. Using a value for frequency is optional; you can use any value if needed(e.g., 50 Hz).]	[6]

Question 2

[20]

For the given *Flash ADC* circuit in Figure 2, $V_3 = -4$ V.

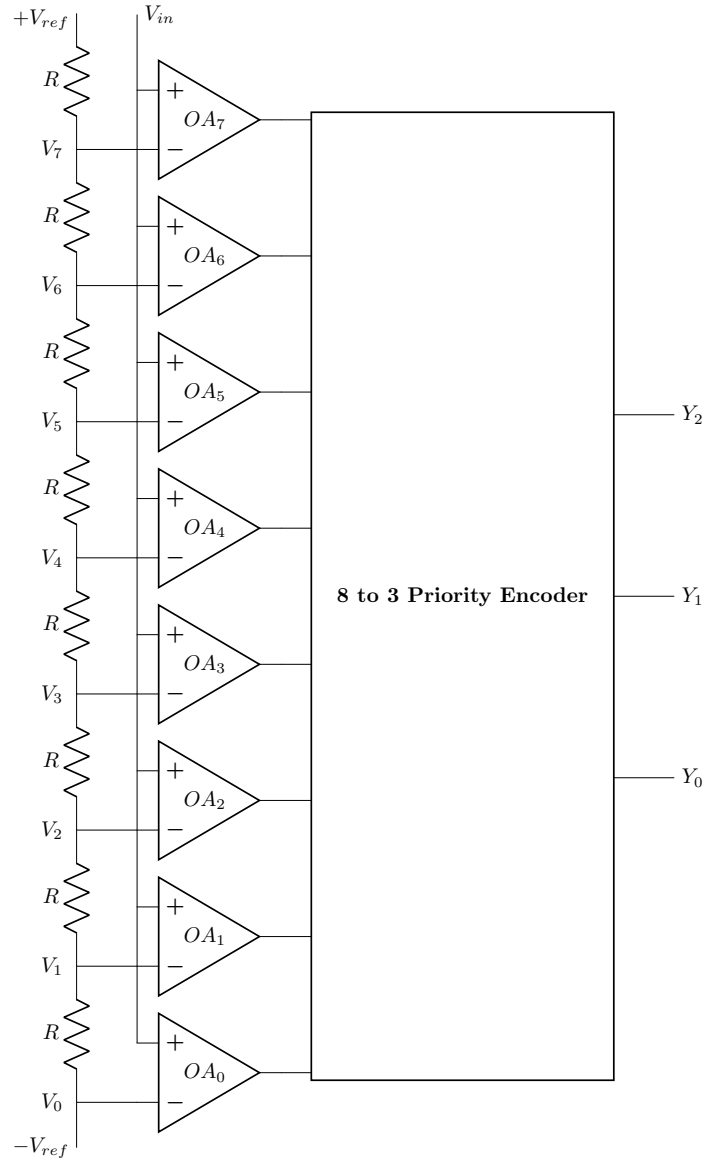


Figure 2: Flash ADC

CO3	(a)	Find the <i>reference voltages</i> and draw the <i>digital output vs input</i> graph.	[7]
CO3	(b)	Now consider a <i>3-bit Binary Weighted DAC</i> with the same value of reference voltage as in (a). If its maximum output voltage matches the <i>highest quantization level</i> of the Flash ADC, and the feedback resistance, $R_f = 2\text{ k}\Omega$, draw the DAC and find the <i>other resistances</i> .	[8]
CO3	(c)	Suppose, a <i>Dual Slope ADC</i> was used in (a) instead, with the <i>same</i> reference voltage. If the counter's output is <i>10110101</i> (in binary) and the clock speed is 1.5 MHz, what is the corresponding <i>input voltage</i> and <i>conversion time</i> ?	[5]

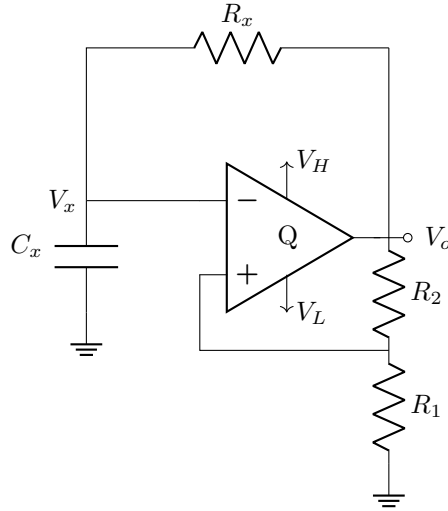


Figure 3: Square Wave Generator

The above circuit generates a square wave with 200 Hz frequency and 40% duty cycle. Here, $R_1 = R_2$, $V_H = +8$ V, $V_L = -8$ V.

CO3	(a)	If the capacitance value is $C_x = 100$ nF, find the value of R_x and the <i>time constant</i> (τ).	[8]
CO3	(b)	Draw the <i>input-output vs time waveform</i> on the same graph with proper labeling on the amplitude and time axis.	[6]

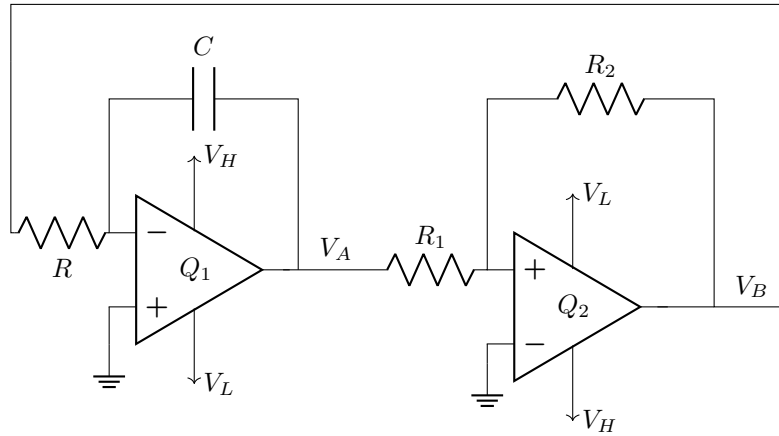


Figure 4: Triangular Wave Generator

In Figure 4, $R_1 = 10$ k Ω , $R_2 = 80$ k Ω .

CO3	(c)	For the above circuit, if $V_L = -V_H$, show that the frequency will be $f = \frac{2}{RC}$. (show proper calculation)	[6]
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Question 4

[20]

In the Figure 5,

Parameter	M_1	M_2	M_3
K_n	$0.3 \frac{mA}{V^2}$	$0.4 \frac{mA}{V^2}$	$0.3 \frac{mA}{V^2}$
V_{TN}	0.6 V	0.7 V	0.6 V

Here, $K_n = \frac{1}{2} \mu_n C_{ox} \frac{W}{L}$

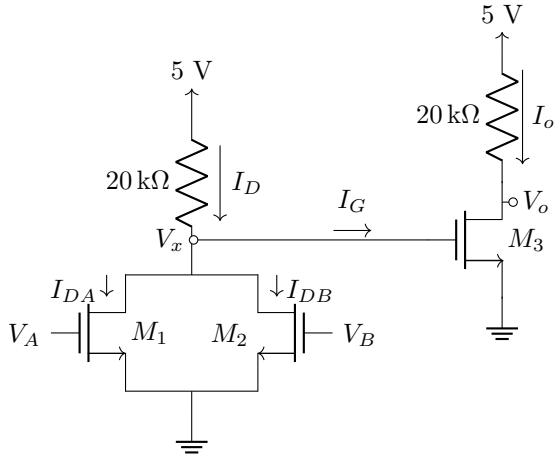


Figure 5: NMOS Circuit

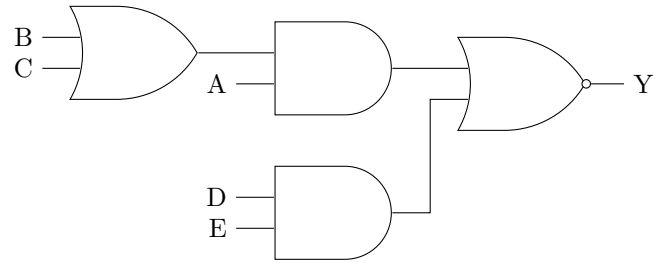


Figure 6: Boolean Logic Circuit

CO1	(a)	From the above Figure 5, determine the currents I_D , I_{DA} , I_{DB} , I_o , V_x and V_o for the cases- i) both inputs are high ii) both inputs are low. (Consider 5 V as logic high and 0 V as logic low)	[8]
CO1	(b)	Determine the <i>power dissipations</i> for the same cases as in (a).	[6]
CO1	(c)	Find the <i>boolean expression</i> for output Y from Figure 6 and using that expression, design an equivalent CMOS logic circuit of Figure 6.	[6]